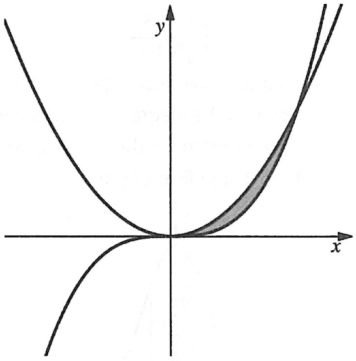
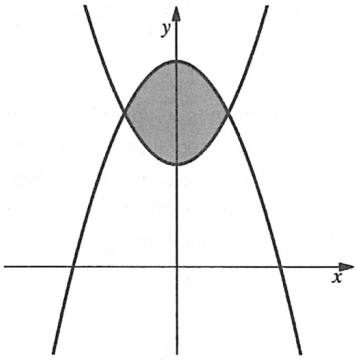
Integration: Areas between Curves

Questions 1–12

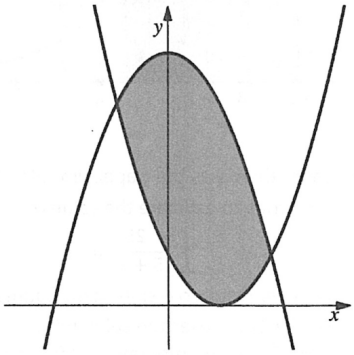
Question 1. The diagram below shows the curves $y = x^2$ and $y = x^3$. Find the exact shaded area between the two curves.



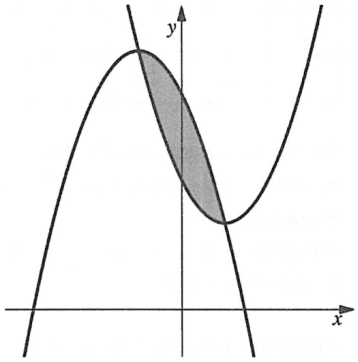
Question 2. The diagram below shows the curves $y = 4 - x^2$ and $y = x^2 + 2$. Find the exact shaded area between the two curves.



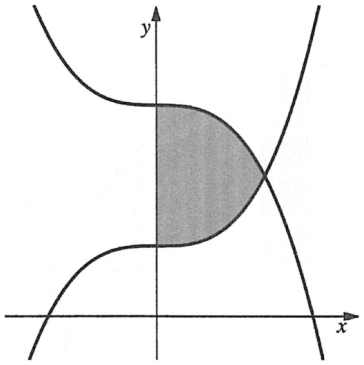
Question 3. The diagram below shows the curves $y = 5 - x^2$ and $y = (x-1)^2$. Find the exact shaded area between the two curves.



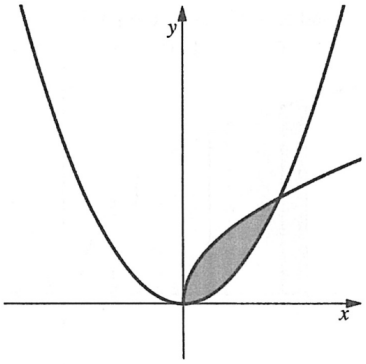
Question 4. The diagram below shows the curves $y = (x-1)^2 + 2$ and $y = 6 - (x+1)^2$. Find the exact shaded area between the two curves.



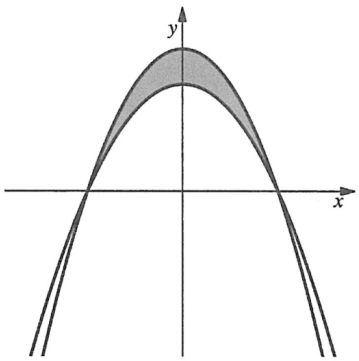
Question 5. The diagram below shows the curves $y = x^3 + 1$ and $y = 3 - x^3$. Find the exact shaded area between the two curves and the y-axis.



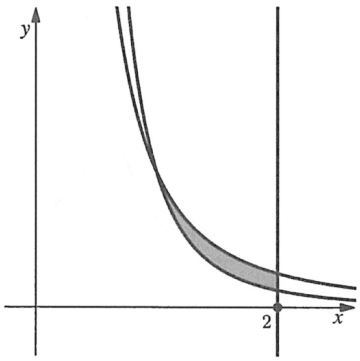
Question 6. The diagram below shows the curves $y = x^2$ and $y = \sqrt{x}$. Find the exact shaded area between the two curves.



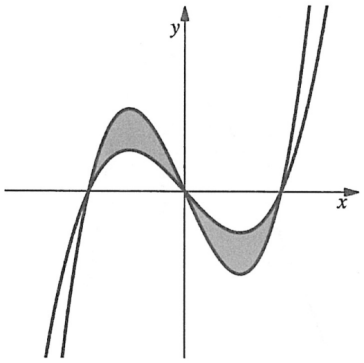
Question 7. The diagram below shows the curves $y = 1 - x^2$ and $y = \frac{4}{3}(1 - x^2)$. Find the exact shaded area between the two curves.



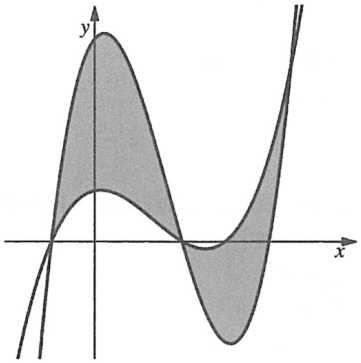
Question 8. The diagram below shows the curves $y = \frac{1}{x^2}$ and $y = \frac{1}{x^3}$, and the line $x = 2$. Find the exact shaded area.



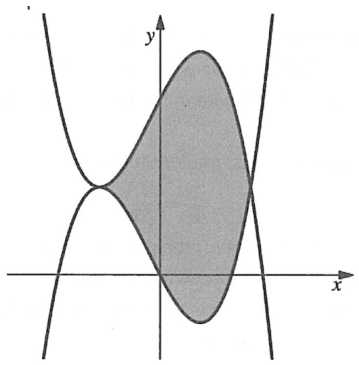
Question 9. The diagram below shows the curves $y = x(x-1)(x+1)$ and $y = 2x(x-1)(x+1)$. Find the exact shaded area between the two curves.



Question 10. The diagram below shows the curves $y = (x+1)(x-2)(x-3)$ and $y = 3(x+1)(x-2)(x-4)$. Find the exact shaded area.



Question 11. It is given that $\mathrm{f}(x) = 8x^3+6x^2-36x$. The diagram below shows the curves $y = \mathrm{f}(x)$ and $y = 81-\mathrm{f}(x)$. The curves intersect at precisely two points. Find the exact shaded area.



Question 12. The diagram below shows the curves $y = \frac{1}{x^2}$ and $y = \frac{2}{x^2}$, and the lines $y = 1$ and $y = 2$. Find the exact shaded area.

